Xiang Zheng

Contact Information

Lau Ming Wai Academic Bldg, 8/F
City University of Hong Kong
Kowloon, Hong Kong SAR, China

Email: xiang.zheng@cityu.edu.hk
Webpage: https://x-zheng16.github.io
GitHub: https://github.com/x-zheng16
Google Scholar: 811 citations, h-index 9

Research Interests

Reinforcement Learning, Trustworthy AI, Agentic AI, Embodied AI

Education

City University of Hong Kong, Hong Kong SAR, China **2020-2024**
Ph.D. in Computer Science
Supervisor: Prof. Cong Wang
Fields: Reinforcement Learning, Trustworthy AI
GPA: 4.00/4.00

Tsinghua University, Beijing, China **2016-2019**
M.S. in Control Science and Engineering
Supervisor: Prof. Tao Zhang
Fields: Reinforcement Learning, Intelligent Robotics
GPA: 3.55/4.00

Beihang University, Beijing, China **2012-2016**
B.S. in Automation, B.S. in Mathematics (Dual Degree)
Selected into the Shen Yuan Honors College (top 0.5%, 1 in 200)
GPA: 3.74/4.00 (222.5 credits, 92 courses)

Research & Work Experience

Hong Kong Institute of AI for Science (HKAI-Sci) **2026-present**
Hong Kong SAR, China
Research Assistant Professor
Topic: Self-evolving science agents and trustworthy AI.
- Research Leader of SciencePal 2.0.

City University of Hong Kong, Hong Kong SAR, China **2024-2026**
Postdoctoral Fellow
Supervisor: Prof. Cong Wang
Topic: Reinforcement learning for trustworthy & embodied AI.
- Step-wise temporal red-teaming of multi-modal toxicity, accepted by ICML'26.
- Curious code agents revealing system prompts in frontier LLMs, accepted by ICML'26.
- Weak-defense leverage for stronger jailbreaks on VLMs, published in Pattern Recognition.

- Lightweight benchmark for RL-based red teaming, published in Frontiers of Computer Science.
- LLM-driven red teaming of text-to-image generators, accepted by CVPR'26.
- Reinforced blue teaming for VLMs against jailbreak attacks, accepted by ICLR'25.
- Curiosity-driven auditing for LLMs, accepted by AAAI'25.

City University of Hong Kong, Hong Kong SAR, China **2020-2024**
Graduate Research Assistant
Supervisor: Prof. Cong Wang
Topic: Efficient exploration strategies for reinforcement learning
- Constrained intrinsic motivation for unsupervised RL, accepted by IJCAI'24.
- Intrinsically motivated adversarial policy for robotic RL agents, accepted by DSN'24.

Xi'an Jiaotong University, Xi'an, China **2019**
Visiting Researcher
Supervised by Prof. Chao Shen & Prof. Xingjun Ma
Topic: Adversarial machine learning

National Institute of Informatics, Tokyo, Japan **2018**
Research Intern
Supervised by Prof. Tetsunari Inamura
Topic: Intelligent robotics

Tsinghua University, Beijing, China **2016-2019**
Graduate Research Assistant
Supervised by Prof. Tao Zhang
Topic: Intelligent space robot

The University of New South Wales, Sydney, Australia **2016**
Research Intern
Supervised by Prof. Elias Aboutanios
Topic: Spacecraft de-orbiting control

Publication

Journal

J-1. Yunhan Zhao, **Xiang Zheng**, Yige Li, Xingjun Ma, "Defense-to-Attack: Bypassing Weak Defenses Enables Stronger Jailbreaks in Vision-Language Models," *Pattern Recognition*, 2026, art. no. 113805.
Impact Factor 7.6, JCR Rank Q1.

J-2. **Xiang Zheng**, Xingjun Ma, Wei-Bin Lee, Cong Wang, "OpenRedRL: A Light-Weight Benchmark for Reinforcement Learning-Based Red Teaming,"

Frontiers of Computer Science, 2026.
Impact Factor 4.6, JCR Rank Q2.

J-3. Yi Liu, **Xiang Zheng**, Chengjun Cai, Xingliang Yuan, Cong Wang, "Adapting Large Language Models for Encrypted Traffic Analysis Services: An Efficient Realization with Mixture of LoRA Experts," *IEEE Transactions on Services Computing*, vol. 19, no. 2, pp. 906-919, 2026.
Impact Factor 5.5, JCR Rank Q1.

J-4. Xingjun Ma, ..., **Xiang Zheng**, et al., "Safety at Scale: A Comprehensive Survey of Large Model and Agent Safety," *Foundations and Trends in Privacy and Security*, vol. 8, no. 3-4, 2026.

J-5. Yuxue Cao, Shengjie Wang, **Xiang Zheng**, Wenke Ma, Xinru Xie, Lei Liu, "Reinforcement Learning With Prior Policy Guidance for Motion Planning of Dual-Arm Free-Floating Space Robot," *Aerospace Science and Technology*, vol. 136, pp. 108098, 2023.
Impact Factor 5.6, JCR Rank Q1 (3/34).

J-6. Shengjie Wang, Yuxue Cao, **Xiang Zheng**, Tao Zhang, "A Learning System for Motion Planning of Free-Float Dual-Arm Space Manipulator Towards Non-cooperative Object," *Aerospace Science and Technology*, vol. 131, pp. 107980, 2022.
Impact Factor 5.6, JCR Rank Q1 (3/34).

J-7. Shengjie Wang, Yuxue Cao, **Xiang Zheng**, Tao Zhang, "Collision-Free Trajectory Planning for a 6-DoF Free-Floating Space Robot via Hierarchical Decoupling Optimization," *IEEE Robotics and Automation Letters* **(RA-L)**, vol. 7, no. 2, pp. 4953–4960, 2022.
Impact Factor 5.2, JCR Rank Q2 (10/30).

Conference (* for co-first author, underlined for corresponding author.)

C-1. **Xiang Zheng**, Yutao Wu, Hanxun Huang, Yige Li, Xingjun Ma, Bo Li, Yu-Gang Jiang, Cong Wang, "Just Ask: Curious Code Agents Reveal System Prompts in Frontier LLMs," in **ICML**, 2026.

C-2. Xutao Mao, Liangjie Zhao, Tao Liu, **Xiang Zheng**, Hongying Zan, Cong Wang, "STARE: Step-wise Temporal Alignment and Red-teaming Engine for Multi-modal Toxicity Attack," in **ICML**, 2026.

C-3. Zilong Wang, **Xiang Zheng**, Xiaosen Wang, Bo Wang, Xingjun Ma, Yu-Gang Jiang, "GenBreak: Red Teaming Text-to-Image Generators Using Large Language Models," in **CVPR**, 2026.

C-4. Yunhan Zhao, **Xiang Zheng**, Lin Luo, Yige Li, Xingjun Ma, Yu-Gang Jiang, "BlueSuffix: Reinforced Blue Teaming for Vision-Language Models Against Jailbreak Attacks," in **ICLR**, 2025.

C-5. **Xiang Zheng**, Longxiang Wang, Yi Liu, Xingjun Ma, Chao Shen, Cong Wang, "CALM: Curiosity-Driven Auditing for Large Language

Models," in **AAAI**, 2025.

C-6. **Xiang Zheng**, Xingjun Ma, Chao Shen, Cong Wang, "Constrained Intrinsic Motivation for Reinforcement Learning," in **IJCAI**, 2024.
Acceptance Rate 791/5651=14.00%

C-7. **Xiang Zheng**, Xingjun Ma, Shengjie Wang, Xinyu Wang, Chao Shen, Cong Wang, "Toward Evaluating Robustness of Reinforcement Learning With Adversarial Policy," in **DSN**, 2024.
Acceptance Rate 42/203=20.69%

C-8. Shengjie Wang, Fengbo Lan, **Xiang Zheng**, Yuxue Cao, Oluwatosin Oseni, Haotian Xu, Tao Zhang, Yang Gao, "A Policy Optimization Method Towards Optimal-Time Stability," in **CoRL**, 2023.

C-9. Shengjie Wang, **Xiang Zheng**, Yuxue Cao, Tao Zhang, "A Multi-target Trajectory Planning of a 6-DoF Free-Floating Space Robot via Reinforcement Learning," in **IROS**, 2021.

C-10. Shengjie Wang, Yuxue Cao, **Xiang Zheng**, Tao Zhang, "An End-to-End Trajectory Planning Strategy for Free-Floating Space Robots," in **CCC**, 2021.

C-11. Shihao Zhao, Xingjun Ma, **Xiang Zheng**, James Bailey, Jingjing Chen, Yu-Gang Jiang, "Clean-Label Backdoor Attacks on Video Recognition Models," in **CVPR**, 2020.
Acceptance Rate 1470/6656 = 22.09%

C-12. **Xiang Zheng**, Ziwei Wang, Tao Zhang, "Robust Finite-Time Attitude Tracking Control for Nonlinear Quadrotor With Uncertainties and Delays," in **ROBIO**, 2017.

Preprint & Submission

1. Xiao Li*, **Xiang Zheng***, ..., Xingjun Ma, Yu-Gang Jiang, "Safety in Embodied AI: A Survey of Risks, Attacks, and Defenses," arXiv preprint arXiv:2605.02900, 2026. (*Equal contribution)
2. Sirui He, Chujie Chen, **Xiang Zheng**, Zhihang Liu, Cong Wang, "Refining Specs For LLM-Based RTL Agile Design," submitted to **ICCAD'26**.
3. Longxiang Wang, **Xiang Zheng**, Xuhao Zhang, Yao Zhang, Ye Wu, Cong Wang, "OptiLeak: Efficient Prompt Reconstruction via Reinforcement Learning in Multi-tenant LLM Services," arXiv preprint arXiv:2602.20595, submitted to **ESORICS'26**.
4. Jiayu Li, Yunhan Zhao, **Xiang Zheng**, Zonghuan Xu, Yige Li, Xingjun Ma, Yu-Gang Jiang, "AttackVLA: Benchmarking Adversarial and Backdoor Attacks on Vision-Language-Action Models," arXiv preprint arXiv:2511.12149, under revision.

5. Yutao Wu, Xiao Liu, Yinghui Li, Yifeng Gao, Yifan Ding, Jiale Ding, **Xiang Zheng**, Xingjun Ma, "ADMIT: Few-shot Knowledge Poisoning Attacks on RAG-based Fact Checking," arXiv preprint arXiv:2510.13842, under revision.
6. Zonghuan Xu, Jiayu Li, Yunhan Zhao, **Xiang Zheng**, Xingjun Ma, Yu-Gang Jiang, "DropVLA: An Action-Level Backdoor Attack on Vision-Language-Action Models," arXiv preprint arXiv:2510.10932.
7. Jiale Ding, **Xiang Zheng**, Yutao Wu, Cong Wang, Wei-Bin Lee, Ling Pan, Xingjun Ma, Yu-Gang Jiang, "RedTopic: Toward Topic-Diverse Red Teaming of Large Language Models," arXiv preprint arXiv:2507.00026, submitted to **TDSC**.
8. Ruofan Wang, **Xiang Zheng**, Xiaosen Wang, Cong Wang, Xingjun Ma, "RedDiffuser: Red Teaming Vision-Language Models for Toxic Continuation via Reinforced Stable Diffusion," arXiv preprint arXiv:2503.06223.
9. Yuxue Cao, Wenhao Zhao, Shengjie Wang, **Xiang Zheng**, Wenke Ma, Ziwei Wang, Tao Zhang, "A Learning-based Adaptive Compliance Method for Symmetric Bi-manual Manipulation," arXiv preprint arXiv:2303.15262.

Projects

Foxconn-CityUHK Joint Research Centre — **2026 – now**
Red Teaming for Intelligent Agents — 500K HKD

Foxconn-CityUHK Joint Research Centre — **2025 – 2026**
Intrinsically Motivated RFT for Red Teaming Black-Box LLMs — 500K HKD

CRF, No. 8730083 — **2023 – now**
Enabling Metadata-Private and Accountable Networks at Scale — 7.52M HKD

NSFC, No. 9054034 — **2022 – now**
Towards Secure and Privacy-enhanced Machine Learning as a Service — 1M CNY

RFS, No. 9062004 — **2022 – now**
Building Privacy-Assured and Scalable Encrypted Databases With Secure Enclave — 5.16M HKD

CAST — **2017 – 2019**
Learning-Based *** *** Control Technology for *** Targets — 1.8M CNY

***** — **2018 – 2019**
Research on Deep Learning Algorithms for Intelligent *** — 600K CNY

Invited Talks

Hong Kong, CityU Joint Summer School on AI for Science — **2026**
Hands-on with Claude Code: Frontier Code Agents for Research and Browser/Computer Use

Hong Kong, HKAI-Sci, CityUHK — **2026**
Build Your Own Self-Evolving Science Agent with SciencePal 2.0

Shenzhen, ByteDance — **2026**
Agentic Red Teaming and Blue Teaming

Hong Kong, College of Computing, CityUHK (InnoCenter) — **2026**
Is OpenClaw All You Need? (Fireside Chat)

Online, Prof. Ziwei Wang's Web-Seminar, Lancaster University — **2026**
From Robot Learning to Self-Evolving OpenClaw: Building and Securing AI Agent Systems

Hong Kong, HKAI-Sci, CityUHK — **2026**
Harness Engineering

Online, Institute of Trustworthy Embodied AI, Fudan (Prof. Xingjun Ma's Group) — **2026**
Reinforcement Learning for Agent Safety Evaluation

Xi'an, CCF YOCSEF — **2025**
Towards Robust Embodied AI: Skill Discovery and Red Teaming

Shenzhen, Southern University of Science and Technology — **2025**
Reinforcement Learning-Based Adversarial Safety Evaluation and Defense Enhancement for Large Language Models

Xi'an, CCF YOCSEF — **2025**
Reinforcement Fine-Tuning for Large Model Red-/Blue-Teaming

Xi'an, Northwestern Polytechnical University — **2024**
Curiosity-Driven Auditing for LLMs

Xi'an, Northwestern Polytechnical University — **2024**
Efficient Intrinsically Motivated Adversarial Policy Learning

Gold Coast, Griffith University — **2024**
Intrinsically Motivated Adversarial Policy

Melbourne Connect, The University of Melbourne — **2024**
Towards Efficient Evasion Attacks Against RL

Awards & Honors	**ICML Gold Reviewer Award (Top 25%)** ICML Program Chairs Reward: Complimentary ICML 2026 registration	**2026**
	IJCAI Travel Grant IJCAI Organization and AIJ Division Amount: USD 500	**2024**
	International Conference Grant City University of Hong Kong Amount: HKD 10,000	**2024**
	DSN Student Travel Grant DSN Student Travel Awards Committee Amount: USD 1,500	**2024**
	Research Activities Fund City University of Hong Kong Amount: HKD 96,000	**2022-2023**
	Institutional Research Tuition Grant City University of Hong Kong Amount: HKD 168,384	**2020-2024**
	CityUHK Presidential PhD Scholarship (2/73) City University of Hong Kong Amount: HKD 1,353,120	**2020-2024**
	NII MOU Research Activities Grant (1/106) National Institute of Informatics, Japan Amount: JPY 68,400	**2018**
	CSC Scholarship China Scholarship Council Amount: AUD 6,400	**2016**
	Stars of Advanced Engineering Shen Yuan Honors College, Beihang University Top-2 of 50 honors students; highest honor at the Honors College.	**2015**
Teaching & Experience	**Teaching Assistant**, City University of Hong Kong CS5293, Topics on Information Security	**Semester B 2023/24**

Teaching Assistant, City University of Hong Kong — **Semester A 2023/24**
CS4394, Information Security and Management

Teaching Assistant, City University of Hong Kong — **Semester B 2021/22**
CS4293, Topics in Cybersecurity
CS5293, Topics on Information Security

Teaching Assistant, City University of Hong Kong — **Semester A 2021/22**
CS4394, Information Security and Management
CS5294, Information Security Technology Management

Teaching Assistant, City University of Hong Kong — **Semester B 2020/21**
CS4293, Topics in Cybersecurity
CS6290, Privacy-enhancing Technologies

Teaching Assistant, City University of Hong Kong — **Semester A 2020/21**
CS2310 & CS2311, Computer Programming

Mentored Students

City University of Hong Kong
Longxiang Wang (PhD), RL, AI for Privacy, Agent Safety
Pinsheng Xu (PhD), Multi-Agent System Safety
Xutao Mao (PhD), RL, VLM Safety, Agent Safety [**ICML'26**]
Chenghao Yao (MS), RL, Unlearning
Shuxuan Lye (MS), GUI Agent Safety
Xiao Hu (MS), RL, Agent-as-Judge

Fudan University
Ruofan Wang (PhD), RL, Red Teaming for VLM
Zilong Wang (PhD), RL, Red Teaming for Text-to-Image Model [**CVPR'26**]
Xiao Li (MS), RL, Multi-Agent System Safety
Yunhan Zhao (MS), RL, Blue Teaming for VLM [**ICLR'25**] → **UIUC PhD '26**
Zixing Chen (MS), RL, VLA Safety
Jiayu Li (MS), VLA Safety
Jiale Ding (UG), RL, LLM Safety → **HKUST PhD '26 (HK PhD Fellowship)**
Zonghuan Xu (UG), VLA Safety, Misinformation

Tsinghua University
Shengjie Wang (MS), RL, Safe RL, Robot Learning, Space Robot Control → **Tsinghua PhD '22**

Institute of Automation, CAS
Yuxue Cao (MS), RL, Robot Learning, Space Robot Control → **Siemens**

Northwestern Polytechnical University

Siyuan Zhang (MS), Agent for Crowdsensing

Professional Activity

Program Committee Member
NeurIPS 2026
ICML 2026
ICLR 2025, 2026
CVPR 2026
ECCV 2026
AAAI 2025, 2026
MM 2025, 2026
ICRA 2026

Journal Reviewer
TPAMI, TDSC, TSC, TC, ACM Computing Surveys (CSUR)

External Conference Reviewer
ACL 2026, NeurIPS 2025, ICNP 2025, ESORICS 2022, AsiaCCS 2022, RAID 2021

External Journal Reviewer
IoT-J